Hackathon Rules and Structure

- Team size: 3 to 4 members recommended (minimum 2, maximum 4). Cross-college teams are allowed.
- Duration: 24 continuous hours. All code must be written during the event; use of open-source libraries, public datasets and APIs is permitted and must be declared.
- Each team picks **one** problem statement. Duplicate selections are allowed, since judging is on quality of solution.
- Pre-trained models and AI coding assistants may be used, but teams must be able to explain every part of their solution during evaluation.
- No pre-built projects. Judges may inspect commit history and experiment logs on the shared repository.
- Teams must respect data privacy. Use synthetic or publicly licensed data only; never use real personal or medical data.
- Final submission: a public Git repository with reproducible instructions, an evaluation report (results table plus baseline), a 3-minute demo video and a 6-slide technical pitch.

The Minimum Credible Result Principle

For every problem, the judges ask four questions: (1) Does the core mechanism actually work? (2) Is it compared against a sensible baseline? (3) Are the metrics honest, with no data leakage and no cherry-picking? (4) Do the team members understand why it works and where it fails? Honest negative results with good analysis score higher than inflated claims.

Recommended 24-Hour Timeline

Hours	Phase	What to Achieve
0 - 2	Understand and Plan	Pick problem, lock the MCR and metrics, find datasets, split tasks.
2 - 5	Baseline First	Reproducible environment and a naive baseline with a working evaluation harness.
5 - 15	Core Mechanism	Implement the one hard idea that makes your solution different.
15 - 17	Checkpoint	Mentor review of early results. Cut scope that will not finish.
17 - 21	Evaluate and Improve	Run experiments, ablations, failure analysis; tune.
21 - 23	Report and Demo	Results tables and plots, README, demo video.
23 - 24	Submit and Pitch	Final push, deck, rehearse a 3-minute technical pitch.

Judging Criteria (100 Marks)

Criterion	Marks	What Judges Look For
Technical Depth and Correctness	30	Core mechanism is sound, non-trivial and implemented by the team.
Evaluation Rigor	25	Baselines, proper splits, metrics, ablations, no leakage.
Problem Understanding	10	Grasp of the constraints, assumptions and limitations.
Engineering Quality	15	Reproducibility, code structure, performance, robustness.
Innovation	10	Original technique or insight beyond tutorials.
Presentation and Q&A	10	Clear technical story, honest limitations, strong answers.

Index of Problem Statements

No.	Title	Domain
1	Hallucination-Proof Enterprise RAG with Claim-Level Verification	Advanced AI & Machine Learning
2	Squeezing a Language Model onto a Laptop CPU	Advanced AI & Machine Learning
3	Autonomous Multi-Agent Bug Fixer	Advanced AI & Machine Learning
4	Adversarial Robustness Lab: Attack, Defend, Detect	Advanced AI & Machine Learning
5	Federated Learning with Differential Privacy on Non-IID Clients	Advanced AI & Machine Learning
6	Chest X-Ray Triage with Calibrated Uncertainty and Abstention	Advanced Healthcare & Bioinformatics
7	ICU Sepsis Early-Warning Under Missing Data	Advanced Healthcare & Bioinformatics
8	Real-Time ECG Arrhythmia Detection on a Microcontroller-Class Device	Advanced Healthcare & Bioinformatics
9	Privacy-Preserving Patient Record Linkage Across Hospitals	Advanced Healthcare & Bioinformatics
10	Graph Neural Network for Drug-Target Binding Prediction	Advanced Healthcare & Bioinformatics
11	Grid Load Forecasting with Optimal Battery Dispatch	Energy, Climate & Earth Observation
12	Satellite Change Detection for Flood and Deforestation Mapping	Energy, Climate & Earth Observation
13	Reinforcement-Learning Digital Twin for Precision Farming	Energy, Climate & Earth Observation
14	Peer-to-Peer Energy Trading with Double-Auction Market Clearing	Energy, Climate & Earth Observation
15	Gas Leak Source Localisation from Sparse Sensors	Energy, Climate & Earth Observation
16	Real-Time Fraud Ring Detection on Transaction Graphs	Advanced FinTech & Markets
17	Limit Order Book Simulator and Market-Making Agent	Advanced FinTech & Markets
18	Explainable and Fair Credit Decisions with Counterfactual Recourse	Advanced FinTech & Markets
19	Cross-Border Payment Routing Under Fees, FX and Compliance	Advanced FinTech & Markets
20	Anti-Money-Laundering Network Analyser	Advanced FinTech & Markets
21	Deep Knowledge Tracing with an Adaptive Curriculum	Advanced Education Technology
22	Programming Tutor with Misconception Detection and Hint Ladder	Advanced Education Technology
23	Code Plagiarism Detector Robust to Obfuscation	Advanced Education Technology
24	Live Lecture Intelligence Engine	Advanced Education Technology
25	Rubric-Aligned Short-Answer Grader with Human Routing	Advanced Education Technology
26	Static Malware Family Classifier with Drift Handling	Advanced Cybersecurity (Defensive)
27	Streaming Intrusion Detection Under Concept Drift	Advanced Cybersecurity (Defensive)
28	Zero-Trust Access Gateway with Continuous Risk Scoring	Advanced Cybersecurity (Defensive)
29	Smart-Contract Vulnerability Scanner	Advanced Cybersecurity (Defensive)
30	Attack Reconstruction from System Logs Using Provenance Graphs	Advanced Cybersecurity (Defensive)
31	Multi-Agent RL Traffic Signal Control Across a City Grid	Smart Systems, Robotics & IoT
32	Multi-Camera Tracking with Privacy by Design	Smart Systems, Robotics & IoT
33	Drone Swarm Search-and-Rescue Planner	Smart Systems, Robotics & IoT
34	Building HVAC Digital Twin with Model Predictive Control	Smart Systems, Robotics & IoT
35	LoRa Mesh Disaster Network with a Custom Routing Protocol	Smart Systems, Robotics & IoT
36	Build Raft From Scratch and Break It With Chaos Testing	Distributed Systems, Cryptography & Blockchain
37	Zero-Knowledge Credential Verification	Distributed Systems, Cryptography & Blockchain
38	Cross-Chain Atomic Swap with Failure Recovery	Distributed Systems, Cryptography & Blockchain

No.	Title	Domain
39	Offline-First Replicated Store with CRDTs	Distributed Systems, Cryptography & Blockchain
40	Front-Running Resistant Decentralised Exchange	Distributed Systems, Cryptography & Blockchain
41	Continuous Sign Language Recognition at Sentence Level	Accessibility, Language & Societal Challenges
42	Machine Translation for a Low-Resource Indian Language	Accessibility, Language & Societal Challenges
43	Deepfake Detection That Generalises Across Datasets	Accessibility, Language & Societal Challenges
44	Crisis Informatics: Real-Time Geo-Triage with Credibility Scoring	Accessibility, Language & Societal Challenges
45	Webcam Eye-Gaze Typing for Motor-Impaired Users	Accessibility, Language & Societal Challenges
46	Mini Distributed SQL Query Engine	Systems, Compilers & Research Frontiers
47	Tiny Language Compiler with Optimisation Passes and WebAssembly Backend	Systems, Compilers & Research Frontiers
48	Causal Root-Cause Analysis for Microservice Performance Regressions	Systems, Compilers & Research Frontiers
49	Vehicle Routing with Time Windows: Quantum-Inspired vs Classical Solvers	Systems, Compilers & Research Frontiers
50	Grand Open Challenge: Bring Your Own Research-Grade Problem	Systems, Compilers & Research Frontiers

Domain 1: Advanced AI & Machine Learning

Problem statements 1 to 5

PS-01 Hallucination-Proof Enterprise RAG with Claim-Level Verification

Problem Statement

Standard retrieval-augmented generation still produces confident, unsupported statements. Build a RAG pipeline where every sentence of the final answer is verified against retrieved evidence, and unsupported claims are removed or flagged before the user sees them.

Core Technical Challenges

- Hybrid retrieval (BM25 plus dense) with a cross-encoder reranker
- Sentence-level entailment verification using an NLI model
- Calibrated abstention for unanswerable questions
- Evaluation harness measuring faithfulness, not just fluency

Constraints

Corpus of at least 500 pages; end-to-end latency under 8 seconds per query; no answer sentence without a cited span.

Minimum Credible Result and Metrics

Evaluate on a self-built set of 100 questions (30 deliberately unanswerable). Report faithfulness, answer recall and abstention precision against a naive RAG baseline.

Stretch Goal

Add contradiction detection when two retrieved sources disagree.

PS-02 Squeezing a Language Model onto a Laptop CPU

Problem Statement

Deploy a useful domain-specific language model on a machine with no GPU and under 2 GB of RAM. You must compress, quantize or distill an open model so it stays accurate on a chosen task while running at interactive speed.

Core Technical Challenges

- Post-training quantization versus quantization-aware fine-tuning
- Knowledge distillation from a larger teacher
- KV-cache and memory profiling
- Task-specific accuracy preservation

Constraints

CPU only; memory ceiling 2 GB; model download size under 1.5 GB.

Minimum Credible Result and Metrics

Show a table of size, tokens per second and task accuracy for the original, quantized and distilled models. Target at least 10 tokens/s with under 5 percent accuracy drop.

Stretch Goal

Speculative decoding with a tiny draft model to improve throughput.

PS-03 Autonomous Multi-Agent Bug Fixer

Problem Statement

Given a failing repository and an issue description, build a team of cooperating AI agents that reproduce the bug, localise the faulty code, write a patch and prove it works by running tests in a sandbox.

Core Technical Challenges

- Repository navigation and fault localisation
- Safe sandboxed execution with timeouts
- Patch generation with test-based verification loop
- Agent coordination and cost control

Constraints

All execution inside containers; limit of 25 LLM calls per issue.

Minimum Credible Result and Metrics

Evaluate on 10 issues from SWE-bench Lite or a self-seeded bug set. Report resolve rate, average cost per fix and failure analysis.

Stretch Goal

Auto-generate a regression test for every fix.

PS-04 Adversarial Robustness Lab: Attack, Defend, Detect

Problem Statement

Neural networks can be fooled by tiny, invisible perturbations. Build an end-to-end lab that attacks an image classifier, hardens it, and detects adversarial inputs at inference time.

Core Technical Challenges

- Implement FGSM, PGD and one black-box attack from scratch
- Adversarial training versus input-purification defence
- Detector based on feature-space or uncertainty statistics
- Honest evaluation against adaptive attackers

Constraints

Use CIFAR-10 or a similar dataset; training must complete within 6 GPU-hours total.

Minimum Credible Result and Metrics

Plot clean versus robust accuracy for baseline and defended models across perturbation budgets. Report detector AUROC.

Stretch Goal

Certified robustness via randomized smoothing on a subset.

PS-05 Federated Learning with Differential Privacy on Non-IID Clients

Problem Statement

Hospitals and banks cannot pool raw data. Simulate a federated learning system with heterogeneous clients and formal differential privacy guarantees, and quantify the privacy-utility tradeoff.

Core Technical Challenges

- Non-IID data partitioning across at least 10 clients
- DP-SGD with a privacy accountant (epsilon, delta)
- Gradient compression to reduce communication
- Straggler and dropout handling

Constraints

Simulation on one machine; per-round communication budget enforced.

Minimum Credible Result and Metrics

Curve of accuracy versus epsilon for at least 4 privacy levels, plus bytes communicated per round with and without compression.

Stretch Goal

Membership-inference attack to empirically validate the privacy claim.

Domain 2: Advanced Healthcare & Bioinformatics

Problem statements 6 to 10

PS-06 Chest X-Ray Triage with Calibrated Uncertainty and Abstention

Problem Statement

In radiology, a wrong confident prediction is worse than no prediction. Build a chest X-ray triage model that knows when it does not know and defers to a human, with visual explanations.

Core Technical Challenges

- Multi-label classification on a public chest X-ray dataset
- Calibration (temperature scaling, ECE) and conformal prediction
- Grad-CAM or equivalent localisation sanity checks
- Out-of-distribution input detection

Constraints

Patient-wise train and test split; no data leakage; only public de-identified data.

Minimum Credible Result and Metrics

Report AUROC per finding, expected calibration error, and coverage versus accuracy curve when abstaining.

Stretch Goal

Draft a structured preliminary report from model outputs.

PS-07 ICU Sepsis Early-Warning Under Missing Data

Problem Statement

Sepsis kills quickly and early prediction saves lives, but ICU data is irregular and heavily missing. Predict sepsis hours ahead from streaming vital signs and labs.

Core Technical Challenges

- Irregular time-series modelling and imputation strategy
- Extreme class imbalance and utility-based evaluation
- Prediction horizon of at least 6 hours
- Clinician-friendly explanation of each alert

Constraints

Use PhysioNet Challenge 2019 or MIMIC demo data; strict temporal splits.

Minimum Credible Result and Metrics

Report the PhysioNet utility score, AUROC and alert fatigue rate (false alerts per patient-day) against a logistic regression baseline.

Stretch Goal

Patient-specific early-warning threshold adaptation.

PS-08 Real-Time ECG Arrhythmia Detection on a Microcontroller-Class Device

Problem Statement

Wearables must classify heartbeats locally to save battery and protect privacy. Train an arrhythmia classifier and compress it to run in real time on a Raspberry Pi or equivalent edge device.

Core Technical Challenges

- Signal filtering and R-peak segmentation
- 1D CNN or small transformer with inter-patient split
- Quantization to int8 with minimal accuracy loss
- Latency and energy profiling on target hardware

Constraints

Inference under 50 ms per beat; model under 500 KB.

Minimum Credible Result and Metrics

Report per-class F1 on inter-patient split, model size, latency and a live demo streaming a recorded ECG.

Stretch Goal

On-device anomaly alert with a patient-specific baseline.

PS-09 Privacy-Preserving Patient Record Linkage Across Hospitals

Problem Statement

Two hospitals want to know which patients they share without revealing names or IDs to each other. Build a record-linkage system that tolerates typos and missing fields while leaking minimal information.

Core Technical Challenges

- Bloom-filter or hashing based encoding of identifiers
- Fuzzy matching with blocking for scale
- Privacy analysis against frequency attacks
- Precision and recall trade-off tuning

Constraints

Must scale to 100,000 synthetic records per side; raw identifiers never exchanged.

Minimum Credible Result and Metrics

Report precision, recall and F1 under 0%, 10% and 25% typo noise, plus runtime versus record count.

Stretch Goal

Private set intersection protocol as a stronger privacy mode.

PS-10 Graph Neural Network for Drug-Target Binding Prediction

Problem Statement

Finding which molecule binds which protein is central to drug discovery. Build a GNN-based model that predicts binding affinity, including for molecules and proteins never seen in training.

Core Technical Challenges

- Molecular graph featurisation from SMILES
- Protein sequence encoding and fusion with ligand graph
- Cold-start evaluation splits (new drug, new target)
- Comparison against fingerprint and random-forest baselines

Constraints

Open datasets only (Davis, KIBA, BindingDB subsets).

Minimum Credible Result and Metrics

Report RMSE and concordance index on warm and cold splits, with ablation of the graph component.

Stretch Goal

Attention-based interpretation highlighting binding substructures.

Domain 3: Energy, Climate & Earth Observation

Problem statements 11 to 15

PS-11 Grid Load Forecasting with Optimal Battery Dispatch

Problem Statement

A microgrid operator must predict demand and decide when to charge or discharge a battery to minimise cost. Combine forecasting with mathematical optimisation in a single decision pipeline.

Core Technical Challenges

- Probabilistic load forecasting with weather features
- Mixed-integer or linear program for battery scheduling
- Handling forecast error through rolling-horizon control
- Tariff and degradation modelling

Constraints

Day-ahead schedule must be computed in under 30 seconds.

Minimum Credible Result and Metrics

Report forecast MAPE, cost savings versus a rule-based schedule over a 30-day replay, and battery cycle count.

Stretch Goal

Robust optimisation that accounts for forecast quantiles.

PS-12 Satellite Change Detection for Flood and Deforestation Mapping

Problem Statement

Authorities need rapid maps of what changed on the ground. Process pairs of satellite images and produce pixel-accurate change masks at scale.

Core Technical Challenges

- Co-registration and cloud masking of Sentinel-2 tiles
- Siamese U-Net or transformer change-detection model
- Tiled inference pipeline for large scenes
- Evaluation with IoU and F1 on a labelled benchmark

Constraints

Pipeline must process a 10,000 by 10,000 pixel scene in under 10 minutes on a single GPU.

Minimum Credible Result and Metrics

Report IoU and F1 on OSCD or LEVIR-CD, plus an interactive map comparing before, after and predicted change.

Stretch Goal

Automatic flood extent area estimation with confidence bounds.

PS-13 Reinforcement-Learning Digital Twin for Precision Farming

Problem Statement

Farmers must decide daily how much to irrigate and fertilise under uncertain weather. Build a crop-growth digital twin and train an RL agent that beats expert rules on yield per unit of water.

Core Technical Challenges

- Crop growth simulator (e.g. DSSAT-style or PCSE) wrapped as a Gym environment
- Reward design balancing yield, water and nitrogen runoff
- Training an agent with PPO or SAC across weather seeds
- Robustness to unseen weather years

Constraints

Training must converge within 8 hours of compute.

Minimum Credible Result and Metrics

Compare against threshold-based irrigation on yield, water use and runoff over 50 unseen weather scenarios with confidence intervals.

Stretch Goal

Explain the learned policy as a set of human-readable rules.

PS-14 Peer-to-Peer Energy Trading with Double-Auction Market Clearing

Problem Statement

Households with solar panels can sell surplus power to neighbours. Design and simulate a local market that clears prices fairly while respecting physical grid limits.

Core Technical Challenges

- Double-auction matching algorithm with price discovery
- Network constraints through a simplified power-flow check
- Strategic-bidding agents and manipulation resistance
- Fairness and welfare metrics

Constraints

Simulate at least 100 prosumers over 7 days at 15-minute resolution.

Minimum Credible Result and Metrics

Report social welfare, price stability, congestion violations and comparison with a flat feed-in tariff.

Stretch Goal

Settlement layer on a tamper-evident ledger.

PS-15 Gas Leak Source Localisation from Sparse Sensors

Problem Statement

A handful of sensors around an industrial site report concentrations. Infer where a leak is, and how large, using physics-informed inference rather than guesswork.

Core Technical Challenges

- Gaussian plume forward model with wind uncertainty
- Bayesian inverse problem via MCMC or particle filter
- Optimal sensor placement
- Real-time updating as readings stream in

Constraints

Inference update in under 2 seconds per new reading.

Minimum Credible Result and Metrics

Simulated leak scenarios (at least 200): report localisation error in metres and credible-interval coverage versus a nearest-sensor baseline.

Stretch Goal

Active sensing: recommend where a mobile sensor should move next.

Domain 4: Advanced FinTech & Markets

Problem statements 16 to 20

PS-16 Real-Time Fraud Ring Detection on Transaction Graphs

Problem Statement

Fraudsters operate in coordinated rings that single-transaction models miss. Build a streaming system that detects suspicious subgraphs under strict latency.

Core Technical Challenges

- Streaming graph construction and sliding windows
- Graph features or GNN embeddings updated incrementally
- Scoring latency under 100 ms at 5,000 events per second (simulated)
- Analyst view showing the suspect ring

Constraints

Dataset is synthetic or public (e.g. Elliptic, IEEE-CIS); no real customer data.

Minimum Credible Result and Metrics

Report PR-AUC, p95 latency and throughput, versus a tabular-only baseline.

Stretch Goal

Concept-drift detection that triggers retraining.

PS-17 Limit Order Book Simulator and Market-Making Agent

Problem Statement

Exchanges run on matching engines. Build a limit order book with a price-time priority matching engine and train a market-making strategy that earns spread while controlling inventory risk.

Core Technical Challenges

- Matching engine supporting limit, market and cancel orders
- Realistic order-flow simulator
- Avellaneda-Stoikov or RL-based quoting strategy
- Backtest with transaction costs and risk metrics

Constraints

Engine must process at least 100,000 orders per second in simulation.

Minimum Credible Result and Metrics

Report PnL, Sharpe ratio, maximum inventory and drawdown across 20 simulated market regimes.

Stretch Goal

Adversarial agent that tries to exploit your market maker.

PS-18 Explainable and Fair Credit Decisions with Counterfactual Recourse

Problem Statement

Regulators demand that rejected applicants be told why and what to change. Build a credit model with explanations, actionable counterfactuals and a fairness audit.

Core Technical Challenges

- Accurate baseline model with calibrated probabilities
- Counterfactual generation under feasibility constraints (age cannot decrease)
- Fairness metrics: demographic parity, equalized odds
- Bias-mitigation method and measured trade-off

Constraints

Counterfactuals must change only actionable features.

Minimum Credible Result and Metrics

Report AUROC, disparity before and after mitigation, and the share of rejected applicants given a valid recourse.

Stretch Goal

Group-level recourse cost analysis showing unequal burden.

PS-19 Cross-Border Payment Routing Under Fees, FX and Compliance

Problem Statement

Sending money across countries passes through multiple banks with different fees, delays, liquidity and sanctions rules. Compute optimal multi-path routes under all constraints.

Core Technical Challenges

- Network model with fees, FX spreads, latency and liquidity limits
- Min-cost multi-commodity flow or heuristic routing
- Compliance screening as a constraint
- Dynamic re-routing on link failure

Constraints

Network of at least 200 nodes; route computation under 1 second.

Minimum Credible Result and Metrics

Compare cost, speed and success rate against shortest-path routing across 1,000 simulated payments.

Stretch Goal

Atomic multi-hop settlement with rollback on failure.

PS-20 Anti-Money-Laundering Network Analyser

Problem Statement

Launderers hide money by splitting, layering and cycling it through many accounts. Detect these typologies in large transaction networks and present explainable cases to investigators.

Core Technical Challenges

- Pattern mining for fan-in, fan-out, cycles and scatter-gather
- Graph learning on a highly imbalanced dataset
- Case generation with an auditable evidence trail
- Scalability to millions of edges

Constraints

Use the IBM AML synthetic dataset or equivalent; investigator UI mandatory.

Minimum Credible Result and Metrics

Report precision and recall at fixed investigator workload, versus a rule-based baseline, with three worked case studies.

Stretch Goal

Natural-language case summaries with every claim linked to transactions.

Domain 5: Advanced Education Technology

Problem statements 21 to 25

PS-21 Deep Knowledge Tracing with an Adaptive Curriculum

Problem Statement

Predicting what a student will get right next is the core of personalised learning. Train a knowledge-tracing model and use it to schedule the most useful next exercise.

Core Technical Challenges

- Sequence model (DKT, SAKT or AKT) on student interaction logs
- Mastery estimation per skill with uncertainty
- Exercise selection through a bandit or planning policy
- Offline evaluation of the policy

Constraints

Use ASSISTments or EdNet subsets; chronological splits.

Minimum Credible Result and Metrics

Report AUC for next-response prediction and simulated learning gain of your policy versus random and fixed order.

Stretch Goal

Cold-start handling for brand-new students and questions.

PS-22 Programming Tutor with Misconception Detection and Hint Ladder

Problem Statement

Showing students the solution teaches little. Build a tutor that diagnoses the conceptual mistake behind buggy code and gives progressively stronger hints without revealing the answer.

Core Technical Challenges

- Static and dynamic analysis of student submissions
- Misconception taxonomy and classifier
- Hint generation with an anti-leakage check
- Evaluation with real or realistic student bugs

Constraints

A hint must never contain the full corrected code.

Minimum Credible Result and Metrics

On 50 buggy submissions, report misconception accuracy and hint helpfulness rated by a rubric, plus leakage rate.

Stretch Goal

Personalised hint style based on student history.

PS-23 Code Plagiarism Detector Robust to Obfuscation

Problem Statement

Students rename variables, reorder statements and rewrite loops to hide copying. Build a detector that sees through such transformations and explains its finding.

Core Technical Challenges

- AST and control-flow-graph based similarity
- Handling of semantic (Type-3 and Type-4) clones
- Scalable all-pairs comparison for 1,000 submissions
- Evidence view with aligned code regions

Constraints

Comparison of 1,000 submissions in under 10 minutes.

Minimum Credible Result and Metrics

Create a benchmark of original and obfuscated pairs; report precision, recall and F1 versus token-based tools like MOSS-style n-grams.

Stretch Goal

Detection of AI-generated code stylistic signals, with an honest uncertainty statement.

PS-24 Live Lecture Intelligence Engine

Problem Statement

Turn a live lecture into captions, a growing concept map and an instant question-answering assistant, all in real time and in more than one language.

Core Technical Challenges

- Streaming speech recognition with low latency
- Slide and speech alignment
- Incremental concept-graph extraction
- Grounded Q&A over the lecture so far

Constraints

End-to-end caption delay under 3 seconds; Q&A answer under 5 seconds.

Minimum Credible Result and Metrics

Report word error rate on a sample lecture, caption latency, and concept-graph precision against a human-made graph.

Stretch Goal

Automatic revision quiz generated from the lecture.

PS-25 Rubric-Aligned Short-Answer Grader with Human Routing

Problem Statement

Grading written answers is the biggest teacher workload. Build an automatic grader that follows a rubric, explains each mark and sends uncertain cases to a human.

Core Technical Challenges

- Rubric-conditioned scoring model
- Agreement with human graders measured by quadratic weighted kappa
- Uncertainty estimation for routing
- Bias and consistency checks across answer styles

Constraints

Must route at least 15 percent of answers to humans while maximising accuracy on the rest.

Minimum Credible Result and Metrics

Use ASAP-SAS or a self-labelled set; report QWK, accuracy at each coverage level and per-criterion explanations.

Stretch Goal

Detect answers that game the rubric with keyword stuffing.

Domain 6: Advanced Cybersecurity (Defensive)

Problem statements 26 to 30

PS-26 Static Malware Family Classifier with Drift Handling

Problem Statement

Malware families evolve continuously, so models decay. Build a defensive static-analysis classifier that remains accurate over time and flags samples it has never seen.

Core Technical Challenges

- Feature extraction from PE file metadata (EMBER-style)
- Time-aware train and test splits
- Open-set recognition for unknown families
- Evaluation against feature-space evasion

Constraints

Defensive analysis only on public labelled feature datasets; no live malware handling required.

Minimum Credible Result and Metrics

Report accuracy decay over simulated months, unknown-family detection AUROC and robustness under feature perturbation.

Stretch Goal

Continual-learning update strategy that limits forgetting.

PS-27 Streaming Intrusion Detection Under Concept Drift

Problem Statement

Attack patterns and normal traffic both change. Build an online intrusion-detection system that adapts without retraining from scratch and tells analysts when it is unsure.

Core Technical Challenges

- Streaming feature engineering from flow data
- Drift detectors (ADWIN, DDM) and online learners
- Cross-day generalisation on CIC-IDS2017
- Alert prioritisation and triage UI

Constraints

Process at least 10,000 flows per second in replay mode.

Minimum Credible Result and Metrics

Train on Monday to Wednesday and test on later days; report F1, false-positive rate per hour and adaptation time after drift.

Stretch Goal

Explain each alert with the top contributing flow features.

PS-28 Zero-Trust Access Gateway with Continuous Risk Scoring

Problem Statement

Passwords alone are no longer enough. Build a gateway that evaluates device posture, behaviour and context on every request and dynamically allows, challenges or blocks access.

Core Technical Challenges

- Policy engine with declarative rules
- Behavioural risk model for each session
- Mutual TLS and short-lived credentials
- Step-up authentication flow and audit trail

Constraints

Added latency per request under 20 ms; all decisions logged.

Minimum Credible Result and Metrics

Run a scripted attack-and-normal traffic simulation; report block rate for malicious sessions, false challenge rate and latency overhead.

Stretch Goal

Session-hijack detection through behavioural drift.

PS-29 Smart-Contract Vulnerability Scanner

Problem Statement

Smart-contract bugs have cost billions. Build a defensive static analyser that finds common vulnerability classes and explains the attack path to the developer.

Core Technical Challenges

- Control-flow and data-flow analysis on Solidity or EVM bytecode
- Detection of reentrancy, unchecked calls and access-control flaws
- Reducing false positives with symbolic or heuristic checks
- Developer-friendly reports with fix suggestions

Constraints

Analyse only public benchmark contracts such as SmartBugs and a self-written vulnerable test set.

Minimum Credible Result and Metrics

Report precision and recall per vulnerability class compared against Slither on the same benchmark.

Stretch Goal

Auto-generate a Foundry test that demonstrates each finding in a local sandbox.

PS-30 Attack Reconstruction from System Logs Using Provenance Graphs

Problem Statement

After a breach, analysts must reconstruct what happened from millions of log lines. Build a provenance-graph system that traces an intrusion backwards and forwards and maps it to MITRE ATT&CK.

Core Technical Challenges

- Log ingestion into a causal provenance graph
- Graph reduction to cut noise without losing the attack path
- Anomalous-path scoring
- ATT&CK technique mapping and timeline visualisation

Constraints

Use DARPA TC-style data or a self-generated simulated incident in a lab VM.

Minimum Credible Result and Metrics

Report the fraction of true attack nodes recovered, graph size reduction and analyst time-to-insight on three scenarios.

Stretch Goal

Natural-language incident report with links back to log evidence.

Domain 7: Smart Systems, Robotics & IoT

Problem statements 31 to 35

PS-31 Multi-Agent RL Traffic Signal Control Across a City Grid

Problem Statement

Fixed-timer signals waste millions of hours. Train cooperating agents to control signals across a grid of intersections and beat the baseline under changing demand.

Core Technical Challenges

- SUMO simulation environment integration
- Multi-agent RL with parameter sharing and neighbour communication
- Reward shaping for queue length, waiting time and emissions
- Generalisation to unseen traffic patterns

Constraints

Training within 8 hours; evaluation on a held-out demand profile.

Minimum Credible Result and Metrics

Report average waiting time, throughput and CO2 versus fixed-time and actuated baselines across 10 random seeds.

Stretch Goal

Emergency-vehicle priority corridors without hurting overall flow.

PS-32 Multi-Camera Tracking with Privacy by Design

Problem Statement

Tracking people across city cameras is powerful and dangerous. Build a multi-camera tracker that follows objects while anonymising identities on-device.

Core Technical Challenges

- Detection and multi-object tracking in each camera
- Cross-camera re-identification
- On-device face and plate blurring before any data leaves the camera
- Metrics balancing tracking quality and privacy

Constraints

Use public datasets such as MOT17 or DukeMTMC-style data; no real surveillance footage.

Minimum Credible Result and Metrics

Report IDF1 and MOTA, re-ID rank-1 accuracy and a quantified measure of anonymisation effectiveness.

Stretch Goal

Differential-privacy style aggregation of people-counting statistics.

PS-33 Drone Swarm Search-and-Rescue Planner

Problem Statement

After a disaster, a swarm of drones must cover a large area to find survivors despite limited communication and drone failures. Plan and replan coverage dynamically.

Core Technical Challenges

- Coverage path planning on a probability map
- Decentralised coordination with limited communication range
- Failure detection and live task reallocation
- Simulation of sensor noise and battery limits

Constraints

Simulation of at least 10 drones and a 5 km by 5 km area.

Minimum Credible Result and Metrics

Report area coverage over time, mean time-to-find a target and robustness when 30 percent of drones fail, versus lawn-mower baseline.

Stretch Goal

Bayesian probability map updated from victim-detection confidence.

PS-34 Building HVAC Digital Twin with Model Predictive Control

Problem Statement

Buildings waste enormous energy on heating and cooling. Build a thermal digital twin and a controller that cuts energy use while keeping occupants comfortable.

Core Technical Challenges

- Grey-box thermal model identified from data
- Model predictive control with weather and occupancy forecasts
- Comfort constraints and energy price awareness
- Comparison with thermostat baselines

Constraints

Controller must solve within the control interval (under 1 minute).

Minimum Credible Result and Metrics

Report energy saved, comfort violation hours and model prediction error over a simulated 30-day period.

Stretch Goal

Learned model correction using real sensor data.

PS-35 LoRa Mesh Disaster Network with a Custom Routing Protocol

Problem Statement

When cell towers fall, communities need a network that heals itself. Design a low-power mesh routing protocol and prove it survives massive node failure.

Core Technical Challenges

- Routing protocol design for duty-cycled, low-bandwidth links
- Loop prevention and fast route repair
- Message prioritisation for emergency traffic
- Large-scale simulation plus a small hardware demo

Constraints

Simulate 100 or more nodes; hardware demo with at least 3 ESP32 or LoRa boards if available.

Minimum Credible Result and Metrics

Report packet delivery ratio, latency and overhead when 20 percent, 40 percent and 60 percent of nodes fail, versus flooding.

Stretch Goal

Store-and-forward for disconnected islands.

Domain 8: Distributed Systems, Cryptography & Blockchain

Problem statements 36 to 40

PS-36 Build Raft From Scratch and Break It With Chaos Testing

Problem Statement

Consensus is the hardest part of distributed systems. Implement the Raft algorithm yourself, then try to break it with network partitions, crashes and message delays.

Core Technical Challenges

- Leader election, log replication and safety
- Persistent state and crash recovery
- Fault-injection framework (partition, delay, drop, clock skew)
- Linearizability checking of recorded histories

Constraints

No existing Raft library may be used for the core protocol.

Minimum Credible Result and Metrics

Run at least 1,000 randomised chaos schedules and show zero safety violations; report availability and recovery time.

Stretch Goal

Dynamic membership change and log compaction.

PS-37 Zero-Knowledge Credential Verification

Problem Statement

Prove that you are over 18, or a registered student, without revealing your birth date or identity. Build a working zero-knowledge proof system for selective disclosure.

Core Technical Challenges

- Circuit design for the statement being proven
- Proof generation and verification in the browser or a service
- Issuer signature verification inside the proof
- Performance and security analysis

Constraints

Proof generation under 10 seconds on a laptop; verification under 1 second.

Minimum Credible Result and Metrics

Demonstrate three credentials end to end and report constraint counts, proof times and sizes.

Stretch Goal

Revocation without linking a user across sessions.

PS-38 Cross-Chain Atomic Swap with Failure Recovery

Problem Statement

Two parties on different blockchains want to exchange assets without trusting each other or an exchange. Implement hash-time-locked swaps and prove they handle every failure mode.

Core Technical Challenges

- Hash time-lock contracts on two chains or networks
- Timeout design to prevent loss of funds
- Monitoring and automatic refund
- Adversarial scenario testing (a party disappears, front-running)

Constraints

Use two local test networks or testnets only.

Minimum Credible Result and Metrics

Show successful swaps and refund paths for at least five failure scenarios, with transaction logs.

Stretch Goal

Support for a non-EVM chain such as Bitcoin testnet.

PS-39 Offline-First Replicated Store with CRDTs

Problem Statement

Field workers lose connectivity for days. Build a replicated data store whose copies always converge when they reconnect, without central coordination.

Core Technical Challenges

- Implementation of at least three CRDT types (counter, set, text or map)
- Anti-entropy and efficient delta synchronisation
- Property-based tests proving convergence
- Partition simulation and conflict visualisation

Constraints

Convergence must hold under arbitrary message reordering and duplication.

Minimum Credible Result and Metrics

Run randomised partition tests with at least 5 replicas and report convergence rate, sync bandwidth and merge latency.

Stretch Goal

Collaborative text editing demo built on your CRDT.

PS-40 Front-Running Resistant Decentralised Exchange

Problem Statement

Traders on automated market makers lose money to sandwich attacks. Design and simulate an exchange mechanism that protects ordinary users.

Core Technical Challenges

- Constant-product AMM baseline
- Commit-reveal or frequent batch auction mechanism
- Attacker simulation measuring extracted value
- Economic analysis of liquidity provider impact

Constraints

Simulation on a local chain or a faithful off-chain model.

Minimum Credible Result and Metrics

Report average user slippage loss and attacker profit with and without your mechanism over 1,000 simulated blocks.

Stretch Goal

Encrypted mempool design using threshold encryption (conceptual prototype).

Domain 9: Accessibility, Language & Societal Challenges

Problem statements 41 to 45

PS-41 Continuous Sign Language Recognition at Sentence Level

Problem Statement

Recognising single letters is easy; understanding real sentences is not. Build a pose-based model that translates sequences of signs into text for a restricted but meaningful vocabulary.

Core Technical Challenges

- Pose and hand landmark extraction pipeline
- Temporal model (transformer or CTC-based) for continuous signing
- Handling of co-articulation and signer variation
- Evaluation by word error rate on unseen signers

Constraints

Vocabulary of at least 50 signs; real-time inference on a laptop.

Minimum Credible Result and Metrics

Report word error rate on held-out signers and live demo of at least 10 full sentences.

Stretch Goal

Text-to-sign avatar for two-way communication.

PS-42 Machine Translation for a Low-Resource Indian Language

Problem Statement

Many languages have very little parallel data. Build a translation system for a low-resource language that works offline on a modest device.

Core Technical Challenges

- Data collection and cleaning from open sources
- Back-translation and transfer from related languages
- Fine-tuning a multilingual model with parameter-efficient methods
- Human and automatic evaluation (chrF, BLEU)

Constraints

Model must run offline in under 2 GB memory.

Minimum Credible Result and Metrics

Report chrF and BLEU against a strong multilingual baseline and a human adequacy rating on 50 sentences.

Stretch Goal

Speech-in, speech-out translation on a phone.

PS-43 Deepfake Detection That Generalises Across Datasets

Problem Statement

Most deepfake detectors collapse on fakes from unseen generators. Build a detector and measure, then reduce, the cross-dataset performance drop.

Core Technical Challenges

- Face-forensics model on one dataset
- Cross-dataset testing on at least two others
- Frequency-domain and temporal-consistency features
- Calibrated confidence and failure analysis

Constraints

Use only public research datasets such as FaceForensics++ and Celeb-DF.

Minimum Credible Result and Metrics

Report in-domain versus cross-domain AUC and show the improvement from your generalisation technique.

Stretch Goal

Audio-visual lip-sync inconsistency detector.

PS-44 Crisis Informatics: Real-Time Geo-Triage with Credibility Scoring

Problem Statement

During disasters, social posts flood in with real needs and many rumours. Build a pipeline that classifies, geolocates and clusters reports, scores their credibility and presents priorities to responders.

Core Technical Challenges

- Classification of needs (food, rescue, medical) in noisy multilingual text
- Geoparsing of place names and coordinates
- Event clustering and duplicate removal
- Rumour and credibility scoring

Constraints

Use CrisisNLP/CrisisMMD or replay streams; throughput of at least 100 posts per second.

Minimum Credible Result and Metrics

Report classification F1, geolocation accuracy within 5 km and the precision of top-20 priority clusters.

Stretch Goal

Image-text multimodal damage assessment.

PS-45 Webcam Eye-Gaze Typing for Motor-Impaired Users

Problem Statement

People who cannot use their hands still need to communicate. Build an eye-gaze typing system using only an ordinary webcam, with prediction to make typing fast.

Core Technical Challenges

- Gaze estimation with fast per-user calibration
- Dwell and blink-based selection with error handling
- Language-model next-word prediction to cut keystrokes
- Usability study measuring words per minute

Constraints

Runs on a standard laptop webcam in real time.

Minimum Credible Result and Metrics

Report gaze error in degrees, typing speed in words per minute and error rate with at least 3 test users.

Stretch Goal

Environment control (lights, messages) through gaze commands.

Domain 10: Systems, Compilers & Research Frontiers

Problem statements 46 to 50

PS-46 Mini Distributed SQL Query Engine

Problem Statement

Databases are among the most complex software ever written. Build a small SQL engine that parses a query, optimises it, and executes it in parallel across partitions.

Core Technical Challenges

- SQL subset parser and logical plan
- Cost-based optimisation (join order, predicate pushdown)
- Partitioned parallel execution with shuffle joins
- Benchmarking on TPC-H style queries

Constraints

Must support SELECT, WHERE, JOIN, GROUP BY and ORDER BY.

Minimum Credible Result and Metrics

Run at least 5 TPC-H queries on 1 GB of data; report speedup from 1 to 4 workers and effect of the optimiser.

Stretch Goal

Vectorised execution and basic query caching.

PS-47 Tiny Language Compiler with Optimisation Passes and WebAssembly Backend

Problem Statement

Design a small statically-typed language and build a real compiler for it, including optimisation passes and code generation to run in the browser.

Core Technical Challenges

- Lexer, parser and type checker with good error messages
- Intermediate representation in SSA form
- At least three optimisation passes (constant folding, dead-code elimination, inlining)
- WebAssembly code generation

Constraints

No parser generators for the type checker and optimiser; they must be hand written.

Minimum Credible Result and Metrics

Compile and run five benchmark programs; report speedup from optimisations and correctness against a reference interpreter.

Stretch Goal

A simple register-allocation or loop-optimisation pass.

PS-48 Causal Root-Cause Analysis for Microservice Performance Regressions

Problem Statement

When a distributed application slows down, thousands of metrics change at once. Build a system that finds the true root cause service from traces and metrics, not just correlated symptoms.

Core Technical Challenges

- Service dependency graph construction from traces
- Anomaly detection across metrics, logs and traces
- Causal or random-walk based root-cause ranking
- Fault injection to create ground truth

Constraints

Use a demo microservice benchmark such as Sock Shop or Train Ticket.

Minimum Credible Result and Metrics

Inject at least 30 faults; report Top-1 and Top-3 root-cause accuracy and mean time to localise versus a correlation baseline.

Stretch Goal

Automatic remediation suggestion with rollback safety check.

PS-49 Vehicle Routing with Time Windows: Quantum-Inspired vs Classical Solvers

Problem Statement

Delivery routing under customer time windows is NP-hard. Implement and compare several solvers, including a quantum-inspired approach, on standard benchmarks.

Core Technical Challenges

- Formulate VRPTW and a QUBO encoding
- Adaptive large-neighbourhood search and simulated annealing
- Comparison with OR-Tools on Solomon benchmark instances
- Fair runtime-versus-quality analysis

Constraints

Time limit of 60 seconds per instance for every solver.

Minimum Credible Result and Metrics

Report gap to best-known solutions on at least 15 Solomon instances and a statistical comparison of methods.

Stretch Goal

Dynamic re-optimisation when new orders arrive mid-day.

PS-50 Grand Open Challenge: Bring Your Own Research-Grade Problem

Problem Statement

Teams may propose their own problem in any domain, but it must be genuinely hard: it needs a clear hypothesis, a baseline to beat and a measurable benchmark. The proposal must be approved by a mentor within the first two hours.

Core Technical Challenges

- One-page brief: problem, hypothesis, baseline, benchmark and risks
- Evidence that the problem cannot be solved by wiring together existing APIs
- A rigorous evaluation plan with clear success criteria
- Mentor sign-off before coding begins

Constraints

Proposals without a measurable baseline will be rejected.

Minimum Credible Result and Metrics

A working system plus quantitative results that show improvement over the stated baseline.

Stretch Goal

A short write-up in research-paper style.

Final Guidance from the Professor

Before you begin coding, ask yourself five questions:

- What is the single hardest idea in this problem, and have I started on it before hour 5?
- What is my baseline, and how will I know I beat it?
- Could my numbers be inflated by data leakage or an unfair split?
- Where will my solution fail, and have I measured it?
- If compute, API or internet fails during the demo, do I have cached results to show?

Submission Checklist

[ ] Public repository with reproducible README (setup, data, commands to rerun results)
[ ] Evaluation report: baseline, your method, metrics table, ablation and failure analysis
[ ] Demo video (3 minutes) or live demo link
[ ] 6-slide technical pitch: Problem, Approach, Architecture, Results, Limitations, Next Steps
[ ] List of datasets, models, APIs and open-source tools used
[ ] Team member names and roles

Remember: a small result that is rigorously measured beats a grand claim that is not. Good luck, and enjoy the 24 hours.

TECHNOVA 2026 - Build. Break. Believe.